Research Project (Biology L8)

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Research Project (Biology L8) (A13861)

Short Title: Research Project (Biology L8)
Faculty Science and Computing
Department Science
Cluster Placement and Projects
Author ROMAHONY
Credits 10 **Level:** Advanced

Description of Module / Aims

This module consists of a research and development project, which will give students the facility to plan and carry out a programme of work based significantly on their own initiative.

Programmes

PROJ-0017	BSc (Hons) in Applied Biology with Quality Management (WD_SQMBI_B)	4 / 2 / M
PROJ-0017	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 8 / M

Indicative Content

- Projects will be based on areas such as: developing and validating analytical procedures, environmental analysis, cell culture technology, molecular biology, microbiology, enzymology, applied immunology, computational biology and bioinformatics.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Plan, organize and carry out a research project.
2. Formulate and test hypotheses.
3. Select advanced scientific skills to generate and analyse data and diagnose and troubleshoot technical problems.
4. Develop a personal work plan and accept responsibility for own work.
5. Develop independent thinking and scientific critical judgement to make effective decisions, including a critical reflection of the project in the context of green practices and institute, national and global sustainability initiatives.
6. Formulate effective communicate skills to disseminate scientific information in a variety of forms including oral presentation.
7. Construct a properly referenced report of the project by sourcing, interpreting and applying appropriate and referenced literature.

Learning and Teaching Methods

- Project work - practical laboratory and online based.
- Workshop on green and sustainable practices in research

Assessment Methods

	Weighing	Outcomes Assessed
Final Project	100%	1,2,3,4,5,6,7

Assessment Criteria

<40%: Inadequate achievement of the learning outcomes.

40%-49%: Will be able to show a basic level of achievement in relation to the key learning outcomes. Relatively poor analytical and communication skills.

50%-59%: Reasonable achievement of learning outcomes. Good analytical and communication skills.

60%-69%: In addition, the student will demonstrate more detailed achievements of project goals including demonstrating an ability to evaluate knowledge acquired.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, predict general trends in behaviour including exceptional behaviour, and critically evaluate and relate topics.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	120	
Independent Learning	150	

Essential Material

- "United Nations Sustainable Development Goals (SDGs)." https://sdgs.un.org/goals
- Jones, A., R. Reed and J. Weyers. _Practical skills in Biology_. 5th Edition. UK: Pearson, 2012.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab